

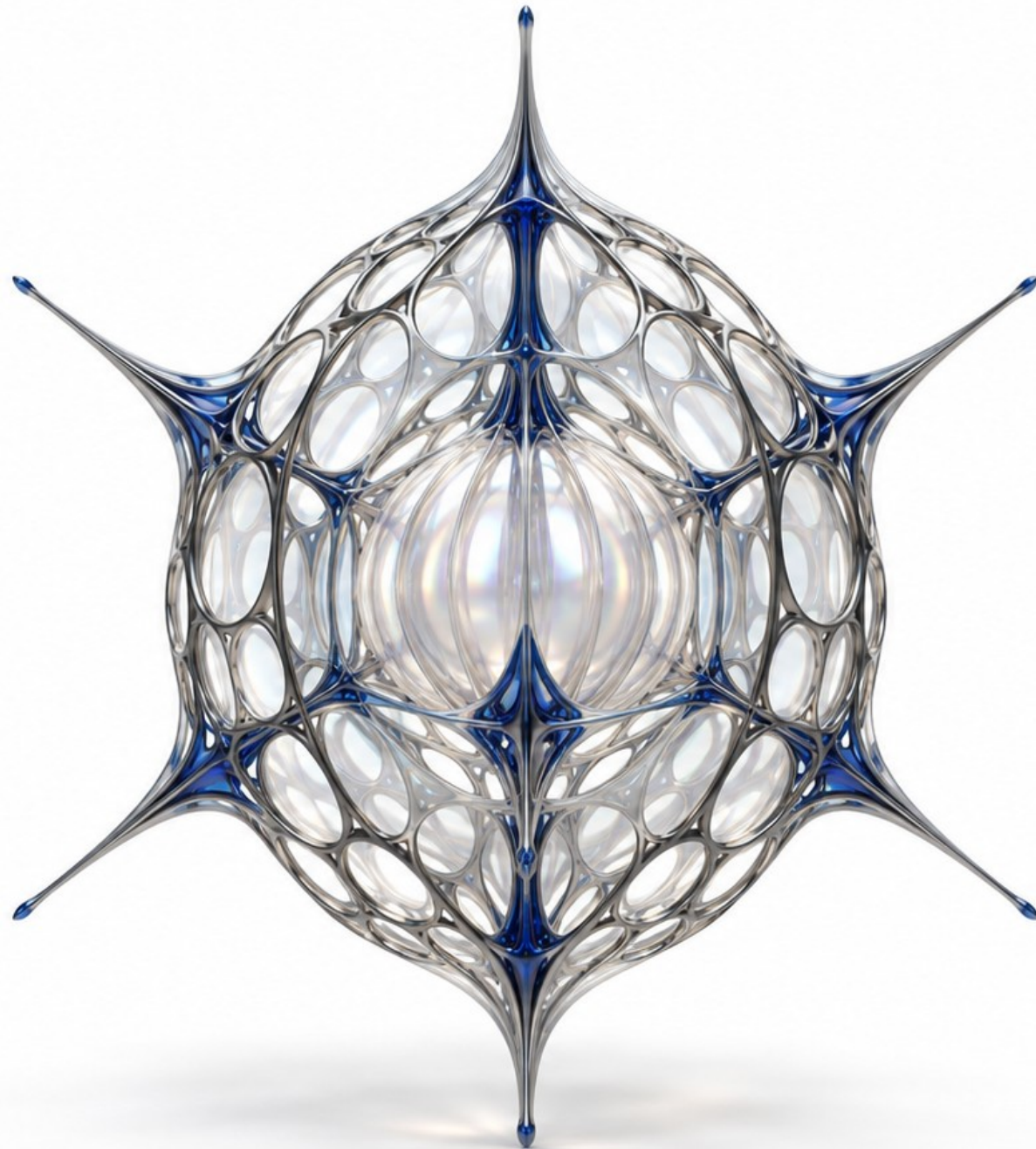
RADIOLARIA OS

Ephemeral Workspace

**A temporary computer
built around a task.**

Semantic work. Current local decisions.
An inspectable execution history.

v0.1 / Main implementation: 14 September 2026



Conceptual artwork

A request becomes a working environment

Requested work

Browse, rate and select photographs.
Adjust exposure and crop. Review video
with audio.

Required boundaries

Keep originals unchanged. Continue visual
work if audio disappears. Save only after
separate confirmation; then close.

Principal run: controlled semantics, real local fixture operations.

Different meaning selects different work

Request	Compiled work	Deferred or omitted
Mixed review	Browse, rate, select, exposure, crop, video, audio	Sidecar write requires separate approval
Photo-only selection	Browse, rate, select	No editing, video or audio operations

Historical EWS3 live roles; completion through strict captured reexecution.

Three semantic duties

Role	Bounded input	Contribution
Orchestrator	Request and declared capabilities	Needs and blocking uncertainty
Semantic architect	Accepted needs and route	Preview, cleanup, save and audio policy
Privacy reviewer	Bounded context and obligations	Allow/block, minimization and conflicts

Principal run: three CONTROLLED_DETERMINISTIC responses; zero provider calls.

The membrane carries bounded evidence

What crosses

Observed semantics, missing evidence, uncertainty, rejected routes, conditions and guards.

What stays separate

Executable commands, private resource handles, local Root decisions and current action authority.

BSEP is evidence, not an ActionCommitPacket, truth or permission.

Root reviews the route and each effect

Semantic acceptance

The route returns to Root with explicit evidence, constraints and unresolved issues.

Current action

Each consequential command receives a finite packet, current Root review and Host/Firewall validation.

Route acceptance does not grant permission to later commands.

The next step consumes the actual output

Work item	Input	Consumed result
compile_contract	Three validated role outputs	Typed command contract
consume_contract	Actual compile_contract output	Session operation gate
Mixed media Work	Actual D-gated media context	Finite frame and PCM plan

Runtime materializes topology; the semantic proposal remains advisory.

A finite set of resource roles

Class	Bounded role
MediaSourceNeedle	Scoped original-source access
RenderEngineNeedle	Derived preview generation
DisplayGatewayNeedle	Derived frames only
ControlSurfaceNeedle	Typed session commands
AudioSinkNeedle	Finite PCM consumption
SidecarWriteNeedle	Independent approval before writing

Recorded contract classes; no current device grants are recreated.

Commands have types, scope and currentness

Finite operations

Browse, rate, select, exposure, crop, playback, pause, seek and media review.

Current boundary

Exact session, version, operation and value. Shell, exec and generic file operations are refused.

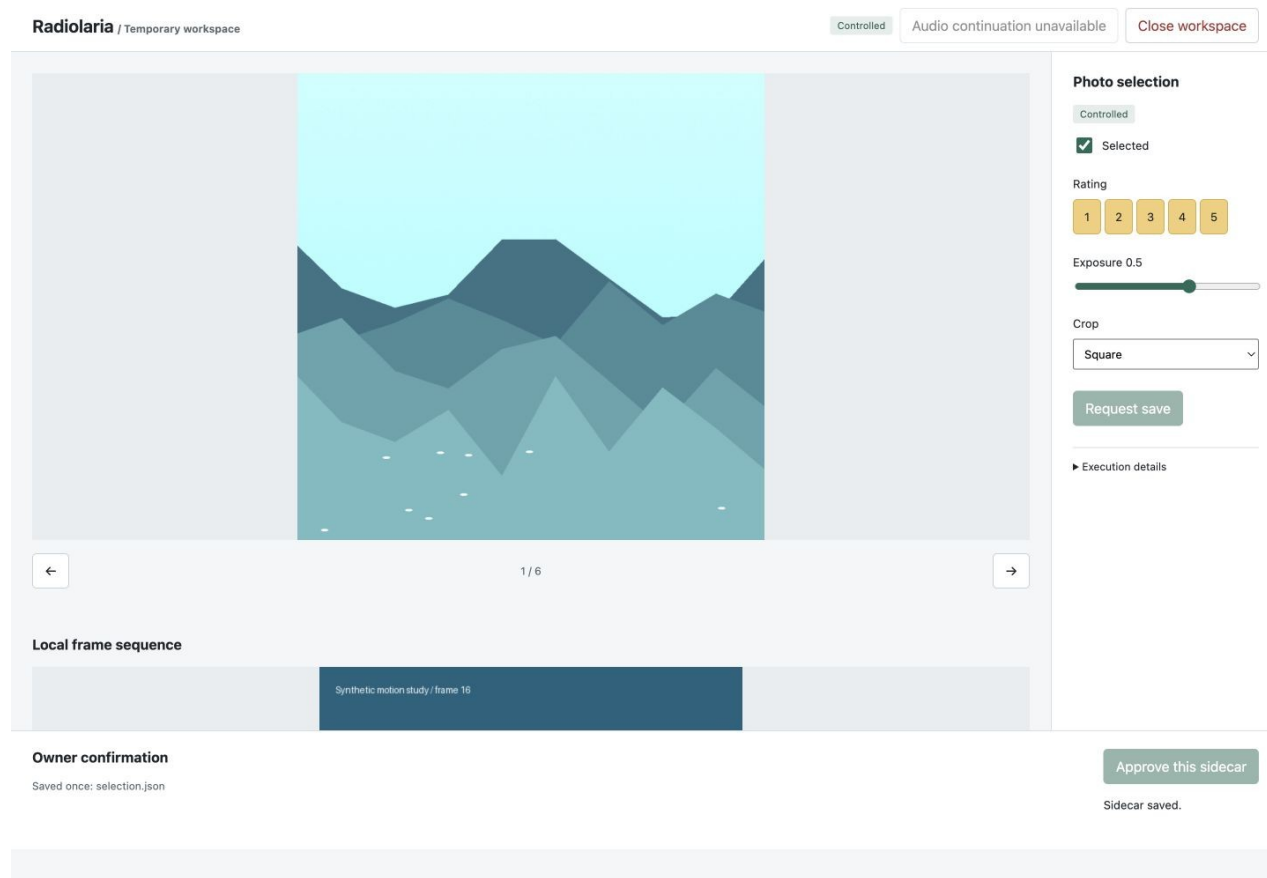
No LLM call, D/E pass or U2 revision per click or frame.

Privacy policy changes the actual transfer

Boundary	Principal permitted fields
Media to session	png, sha256, asset, session
Session to display	png, sha256, frame_ref
Enforcement	Producer, session and display check closed schemas

Metadata minimization; no image-content redaction or OS sandbox claim.

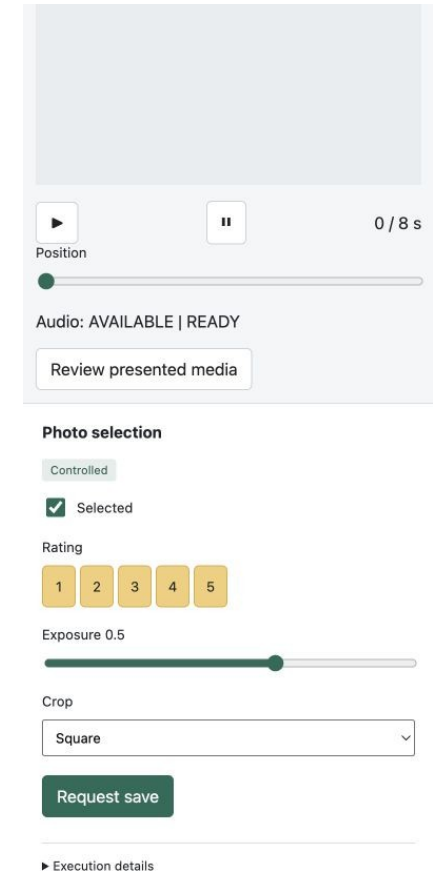
Desktop and mobile operate the same session



Desktop 1440 × 1000

SCRIPTED_BROWSER_TEST. Desktop 1440×1000; mobile 390×844.

RADIOLARIA OS / EPHEMERAL WORKSPACE v0.1



Mobile 390 × 844

AudioSink disappears during real work

4

frames before loss

16,000

PCM samples delivered

Observed

actual child exit

Intentional owned-process failure; synthetic audio, no physical speaker claim.

Fresh evidence keeps old authority unchanged

Event	UTC	Meaning
Old packet expires	11:00:00	Old permission is no longer current
Owned AudioSink dies	11:01:05	Actual resource state changes
New loss evidence acquired	11:01:26	86 seconds after old expiry
New observation expires	11:03:26	Fresh two-minute evidence interval

New evidence window; old packet and Workspace expiry remain unchanged.

Only the affected branch is recomputed

Recompute

The changed audio dependency drives an actual retained D result consumed by E.

Preserve

Unchanged photo Work, preview and unaffected sibling evidence remain bound to their original results.

One top-level D, one public E and one retained D inside E.

Visual work continues without audio

Unchanged

photo Work and preview

+12

video frames, 4 to 16

+0

PCM samples after loss

Optional audio policy SILENT_CONTINUE; no new audio samples.

Saving needs its own approval

Request and review

REQUEST_SAVE forms the exact candidate. Separate owner approval binds the selected content.

Current write

A fresh SAVE packet writes that sidecar once. A changed candidate needs new approval.

One 250-byte sidecar; exact approval; originals preserved.

Memory carries meaning across sessions

Reused meaning

A reviewed recipe supplies bounded needs, obligations and privacy policy through summary-only descent.

Fresh execution

Root review and current resource authority are rebuilt. The durable recipe carries no action permission.

Historical EWS3 LocalDRS; fresh Root, Host, resources and save consent.

Changed sources block the old recipe

Situation	Recorded behavior
Changed source scope	Old recipe: BLOCKED_BY_SCOPE
Fresh reviewed recipe	New candidate eligible; old candidate remains blocked
What was saved?	Bounded summary; zero model calls, effects or activations

Historical DRS evidence; saved-work information does not repeat a write.

Each execution mode keeps its own evidence

Lane	Recorded scope
Historical LIVE	9 attempts; 6 accepted responses; 3 failures
CAPTURED_REEXECUTION	Pinned responses; fresh Host and resources
Principal controlled run	Three semantic responses; real local browser/media
Five local probes	Explicit production boundaries and separate fixtures
Offline replay	Safe evidence verification without runtime execution

No uninterrupted live end-to-end or newly rerun historical proof claim.

Twelve requirements have explicit provenance

12

named requirements

5

current local probes

7

carried historical positions

Bounded proof matrix; not a universal OS or adversarial security certificate.

The workspace closes its owned resources

3

owned service processes
reaped

2

viewer threads closed

Preserved

originals and approved
sidecar

Recorded CLOSED_SUCCESS; dead audio child has no fabricated graceful receipt.

An independent pin anchors safe evidence

Frozen package

18 public files, with source bindings, manifest and integrity records. Historical fields remain unchanged.

Independent verification

The caller supplies the expected pin. Offline replay verifies the package without recreating live authority.

ANCHORED_PASS applies to exact safe derivatives, not hidden private originals.

Recorded times describe different scopes

Stage	Time	Scope
D preparation	113.107 s	Principal mixed setup
Selective E	300.125 s	Actual recomputation
Browser launcher	866.681 s	Complete principal session
EWS4R evidence audit	23.307 s	59 cases
Landing checks	86.395 s	10 narrow tests
Landed public replay	1.375 s	Offline verification

Observed timings; overlapping runtime stages are not additive or a production SLA.

The demonstrated scope stays explicit

Demonstrated

Semantic choices affect work. Authority stays current. Loss is observed. Unaffected results survive. Save and cleanup are evidenced.

Outside this proof

General OS isolation, physical audio, image understanding, production certification and replay of hidden native originals.

Overall EWS acceptance is not self-awarded; native legacy schema limit remains.

Code, evidence and a reader in one showcase

Developer entry points

Inspect the source index and technical appendix. Run portable replay with the independent expected pin.

One-file model reader

Read slide text, exact public evidence and implementation sources with explicit provenance and omissions.

Source 3f8720b. Presentation and XML are explanatory derivatives.